

Class Balanced Sampling for the Training in GANs

Sanghun Kim

Department of Computer Science and Engineering
Kyunghee University, Republic of Korea
powerkei@naver.com

Seungkyu Lee

Department of Computer Science and Engineering
Kyunghee University, Republic of Korea
seungkyu@khu.ac.kr

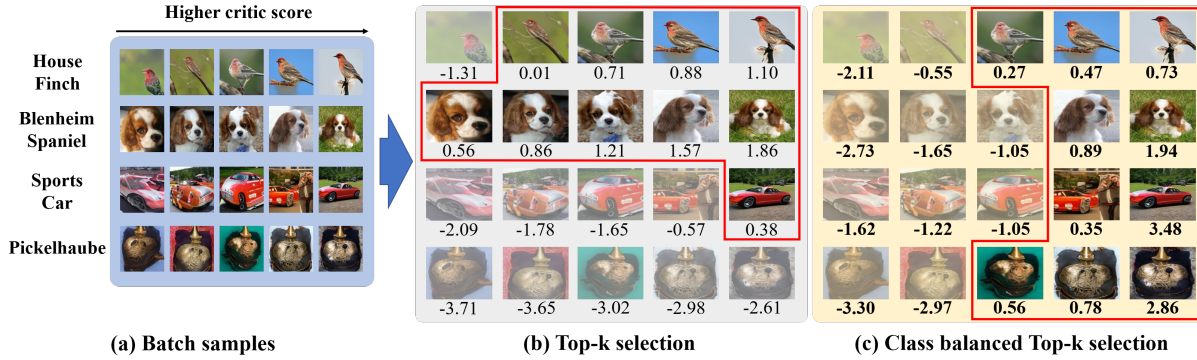


Figure 1: Comparison of Top-k and proposed sampling. (a) Example of batch samples. (b) Example of Top-k selection. Numbers under samples are critic score. Mode collapsed classes with low critic scores may not be selected by Top-k sampling. (c) Example of proposed sampling method. Numbers under samples are Z-scores that is class standardized critic scores.

ABSTRACT

Recently Top-k fake sample selection has been introduced to provide better gradients for training Generative Adversarial Networks. Since the method does not guarantee class balance of selected samples in class conditional GANs, certain classes can be completely ignored in the training. In this work, we propose class standardized critic score based sample selection which enables class balanced sample selection. Our method achieves improved FID score and Intra-FID score compared to prior Top-k selection.

CCS CONCEPTS

• Computing methodologies → Image representations.

KEYWORDS

generative model, adversarial learning, sample selection

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1 INTRODUCTION

GAN(Generative Adversarial Network) is a generative deep neural network model that consists of generator and discriminator networks. The two networks are competitively trained through minimax games with an adversarial loss. Generator makes fake samples from random latent vector representation, then discriminator classifies fake samples and real samples. Even though adversarial learning scheme shows powerful and effective training of the generative network, the stability of learning highly depends on discriminator's performance since generator receives gradients back-propagated from discriminator.

Top-k GAN [Sinha et al. 2020] proposes a sample selection method providing good gradients for the training of generator. It argues that low quality samples produce unexpected negative gradients for the training. They show that Top-k generated sample selection based on critic (discriminator) score improves generator training performance in various baseline networks. Top-k GAN reduces the influence of low quality samples by eliminating them in the generator training process. However, it does not guarantee that Top-k selection includes balanced samples from all classes in class conditional GANs. For example, with ImageNet data set of 1000 object classes, generated samples of animal classes such as dogs, cats, birds and insects have higher quality compared to the samples of other classes. This is because animal classes show higher inter-class similarity so that the generator learns the manifold of real samples more easily than other classes. In this case, Top-k GAN tends to select more animal samples than non-animal samples in the training of generator. As the training proceeded, this imbalance becomes severe. In other words, some classes that are relatively difficult for generator to learn manifold of real samples are hardly

selected during the training. As a result, such classes are gradually ignored in the training.

In this work, we propose class balanced Top-k selection based on a class standardized critic score. To this end, z-score is defined and Top-k selection is conducted based on the score so that selected samples keep inter-class balance. Figure 1 compares the difference in the sample selection of Top-k [Sinha et al. 2020] and proposed sampling methods. Generator trained on ImageNet creates higher quality fake images of animal classes compared to non-animal classes. In figure 1 (b), Top-k selects all samples of class 'Blenheim spaniel' whereas all samples of class 'pickelhaube' are rejected. Figure 1 (c) shows Top-k samples of our method using our class standardized critic score. Our method performs class balanced sample selection including samples of mode collapsed classes. In experimental evaluation, our method shows better convergence in training.

2 PROPOSED METHOD

In the class conditional training of GAN, we assume that critic score distribution of each class is Gaussian. We define z-score s_i of a i -th fake sample as follows.

$$s_i = \frac{D(G(z_i, c_i), c_i) - \mu_{c_i}}{\sigma_{c_i}}, \quad (1)$$

where $D(G(z_i, c_i), c_i)$ is critic score of the fake sample which is generated with class condition c_i . μ_{c_i} and σ_{c_i} indicate mean and variance of class c_i respectively. In the training process, a queue for each class is maintained accumulating the scores of observed fake samples for the extraction of distribution parameters of each class. Based on the class standardized critic score s_i , we perform Top-k sample selection for the training.

One potential problem in maintaining class queue is that the embedded space of discriminator changes over training iterations. If there is a class which does not have generated samples for some iterations, class queue is not updated either. If parameters of discriminator are updated significantly, there is corresponding difference in embedding spaces of new critic score and previously stored score in the queue. To prevent this problem, we apply our method after training becomes stable.

Sample selection in the training of GAN suffers from slow convergence because it uses smaller number of fake samples compared to original GAN. Top-k GAN decays selection hyperparameter k . They initially use full batch size and gradually reduce the number of selected samples as the epoch goes on. For fair comparison with Top-k selection, we also adaptively decay k .

3 EXPERIMENTAL EVALUATION

We use ImageNet 64×64 for experimental evaluation. Baseline network of our experiments is BigGAN [Brock et al. 2018]. To accelerate training speed, we use instance selection [DeVries et al.

Table 1: ImageNet data and sample selection ratio between animal classes and non-animal classes

	ImageNet	Top-k GAN	Class balanced Top-k
Animal	39.7%	57.4% (+17.7)	42.4% (+2.7)
Non-animal	60.3%	42.6% (-17.7)	57.6% (-2.7)

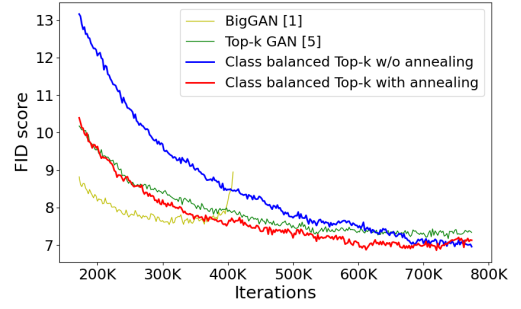


Figure 2: FID scores on 64×64 ImageNet experiments

Table 2: FID and Intra-FID comparison

Method	FID	Intra-FID
Top-k GAN [Sinha et al. 2020]	7.17	65.42(± 16.41)
Class balanced Top-k	6.98	60.69(± 14.41)

2020] which reduces data set by eliminating outlier samples. We run all experiments for 800k iterations. Table 1 shows sample selection ratio between animal classes and non-animal classes. We check selected top-k samples over 1000 iterations with network fixed. Animal classes are classes from 0 to 397 of ImageNet, and non-animal classes are classes from 398 to 999. Top-k selection is more likely to select animal classes because generator learns manifold of animal classes more easily than non-animal classes. Animal and non-animal ratio (57.4 : 42.6) of top-k selection is significantly different from ImageNet data ratio. Top-k selection does not guarantee inter-class balance during training. Our method shows balanced selection ratio (42.4 : 57.6) that is close to actual ImageNet data ratio (39.7 : 60.3). Table 2 compares Fréchet inception distance (FID) scores and Intra-FID scores [Heusel et al. 2017]. We use 10k real and fake samples for calculating FID score. For calculating intra-FID, we use 10k fake samples and 1.3k real samples. This is because ImageNet only has 1.3k samples for each class. Our class balanced sample selection achieves improved FID ($7.17 \rightarrow 6.98$) and Intra-FID ($65.42 \rightarrow 60.69$) scores compared to Top-k selection.

4 CONCLUSION

We propose class balanced Top-k selection for training GAN. Our selection method effectively resolves the inequality problem between classes while maintaining the advantages of Top-k GAN.

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